

AIGR Gene Hypothesis Deep Research — Final Report

Gene: atzA (atrazine chlorohydrolase) | **Organism:** *Pseudomonas* sp. ADP (NCBITaxon:47660) | **UniProt:** P72156 **Focus:** function_assignment | **Hypothesis slug:** function-hypothesis-go-0016810 **Seed hypothesis:** *atzA has hydrolase activity, acting on carbon-nitrogen (but not peptide) bonds (GO:0016810).* **Term under review:** GO:0016810 — hydrolase activity, acting on carbon-nitrogen (but not peptide) bonds | **Evidence type:** IEA | **Original reference:** GO_REF:0000002

Summary

Verdict: REFUTED / OVER-ANNOTATED. The seed hypothesis — that atzA (P72156) has "hydrolase activity, acting on carbon-nitrogen (but not peptide) bonds" (GO:0016810) — is **not supported** by the experimental record and should be **removed** from the gene review. AtzA is an experimentally proven **atrazine chlorohydrolase** (EC 3.8.1.8) that hydrolyzes a **carbon-chlorine (C–Cl) bond**, not a carbon–nitrogen (C–N) bond. The correct, experimentally-supported molecular-function term is **GO:0018788 (atrazine chlorohydrolase activity)**, which sits in a *disjoint* sub-branch of the hydrolase ontology (the "acting on halide bonds" branch, EC 3.8) that does **not** descend from GO:0016810 (the "acting on C–N bonds" branch, EC 3.5).

The strongest evidence is mechanistic and direct. Heavy-water (H₂¹⁸O) isotope-labeling showed the hydroxyl oxygen of the hydroxyatrazine product derives from water, proving AtzA is a **chlorohydrolase (hydrolytic dechlorinase)**, **not an oxygenase** ([PMID:8759853](#)). A near-identical paralog study demonstrated AtzA "exclusively catalyzes dehalogenation of halo-substituted triazine ring compounds and had **no activity with melamine and ammeline**" — i.e., it lacks deaminase (C–N hydrolase) activity entirely ([PMID:11274097](#)). AtzA is a Fe(II)-dependent metalloenzyme that proceeds via a hydrolytic mechanism ([PMID:12450410](#)).

The GO:0016810 annotation is a **single IEA** (ECO:0000256, GO_REF:0000002) that traces to an InterPro fold/domain-superfamily mapping — specifically the IPR011059 "Metal-dependent hydrolase, composite domain superfamily" interpro2go mapping — and not to any experiment

or to the enzyme's EC number. Because AtzA belongs to the amidohydrolase superfamily (a structural fold dominated by C–N-acting deaminases and amidohydrolases), a fold-based electronic pipeline mis-assigned a C–N hydrolase activity term. The EC number that IS assigned to AtzA (EC 3.8.1.8) maps, via the official ec2go mapping, to **GO:0018788**, the correct C-halide term — which already carries **two independent experimental (EXP) annotations** in live GOA. This is a textbook example of fold-based IEA over-annotation landing in the wrong, disjoint ontology branch. **Recommended curation action: remove/down-rank GO:0016810; retain GO:0018788 as the core molecular function.**

Key Findings

F001 — AtzA catalyzes C–Cl bond hydrolysis (dechlorination), not C–N bond hydrolysis

The defining reaction of AtzA is the first, committed step of atrazine catabolism: **atrazine + H₂O → hydroxyatrazine + chloride + H⁺**. This is a hydrolytic dehalogenation — water attacks the ring carbon bearing the chlorine substituent, displacing chloride and installing a hydroxyl. UniProt P72156 records this reaction with EC 3.8.1.8 and an experimentally-supported GO:0018788 (atrazine chlorohydrolase activity, EXP:UniProtKB).

The mechanism was proven directly by isotope labeling. de Souza et al. ([PMID:8759853](#)) purified AtzA and ran the reaction in ¹⁸O-enriched water: *"The purified enzyme in H₂¹⁸O yielded [¹⁸O]hydroxyatrazine, indicating that AtzA is a chlorohydrolase and not an oxygenase."* The appearance of ¹⁸O in the product hydroxyl group establishes that the oxygen comes from solvent water (hydrolysis), not from molecular O₂ (oxygenation). This is the mechanistic crux that places AtzA firmly in the hydrolytic-dehalogenase category.

Critically, the substrate specificity of AtzA excludes C–N hydrolase (deaminase) activity. Seffernick et al. ([PMID:11274097](#)) compared AtzA to its 98%-identical paralog melamine deaminase (TriA) and found that *"AtzA was shown to exclusively catalyze dehalogenation of halo-substituted triazine ring compounds and had no activity with melamine and ammeline."* Melamine and ammeline are deaminated (C–N bond cleavage releasing ammonia) by the paralog, but AtzA does not touch them. UniProt further states AtzA "has no deaminase activity with melamine (PubMed:22768133, PubMed:8759853)." Thus the gene product directly performs C–Cl hydrolysis and directly lacks C–N hydrolysis — the exact opposite of what GO:0016810 asserts.

F002 — GO:0016810 is a disjoint-branch over-annotation (InterPro amidohydrolase-family IEA)

The GO ontology separates hydrolases by the bond they act upon. Using QuickGO `is_a` ancestry traversal:

- **GO:0018788** (atrazine chlorohydrolase activity) ancestry → GO:0016824 (hydrolase acting on acid halide bonds) → GO:0019120 (hydrolase acting on carbon-halide bonds) → GO:0016787 (hydrolase). **GO:0016810 is NOT an ancestor.**
- **GO:0016810** (hydrolase acting on C–N, non-peptide, bonds) ancestry → GO:0016787 (hydrolase) only.

These two terms live in **disjoint sub-branches** of the hydrolase tree — the C-halide branch (corresponding to EC 3.8) versus the C–N branch (corresponding to EC 3.5). An enzyme cannot be correctly annotated to both based on a single activity; they represent chemically distinct bond-cleavage classes. The presence of GO:0016810 on P72156 as an IEA:InterPro annotation is therefore not a more-general parent of the true function — it is a **sibling in the wrong branch**, i.e., a genuine mis-assignment rather than a harmless generalization.

The source of the error is structural homology. AtzA belongs to the **amidohydrolase superfamily** (Pfam PF01979 Amidohydro_1; InterPro IPR006680 Amidohydro-rel), a large (β/α)₈ TIM-barrel metalloenzyme fold whose members are *predominantly* C–N-acting deaminases and amidohydrolases (e.g., cytosine deaminase, adenosine deaminase, urease-related enzymes). AtzA is a rare dehalogenating member of this otherwise C–N-dominated fold. Fold-recognition IEA pipelines, keyed on the superfamily signature, therefore default to the majority function of the fold (C–N hydrolysis) and propagate GO:0016810 — overriding the known, specific dehalogenase chemistry.

F003 — GO:0016810 traces specifically to the IPR011059 composite-domain interpro2go mapping; EC 3.8.1.8 maps to the correct term

Examining the official GO external2go mapping files (current.geneontology.org) pinpoints the exact provenance:

- **ec2go:** `EC:3.8.1.8 → GO:0018788 (atrazine chlorohydrolase activity)`. The EC number assigned to AtzA maps precisely and correctly to the C-halide molecular-function term. If the annotation pipeline had used the EC number, it would have produced the right answer.

- **interpro2go:** IPR011059 (Metal-dependent hydrolase, composite domain superfamily) → GO:0016810. This is the **sole source** of the seed term. The composite metal-dependent-hydrolase fold mapping carries GO:0016810 as its assigned function.
- By contrast, the more specific family signatures **IPR006680 (Amidohydrolase-related)** and **PF01979 (Amidohydro_1)** map only to the generic **GO:0016787 (hydrolase)** — a safe, correct parent that makes no bond-class claim.

The conclusion is that GO:0016810 on P72156 is a **domain-superfamily IEA driven by IPR011059**, not an EC-derived or experiment-derived annotation. It reflects the fold's majority chemistry, not AtzA's actual chemistry. The pipeline that should have governed the MF call — ec2go on EC 3.8.1.8 — points to GO:0018788.

F004 — Live GOA confirms GO:0016810 is a single IEA while GO:0018788 carries two independent EXP annotations

The live QuickGO annotation set for P72156 contains 7 annotations, and their evidence tiers make the curation decision clear-cut:

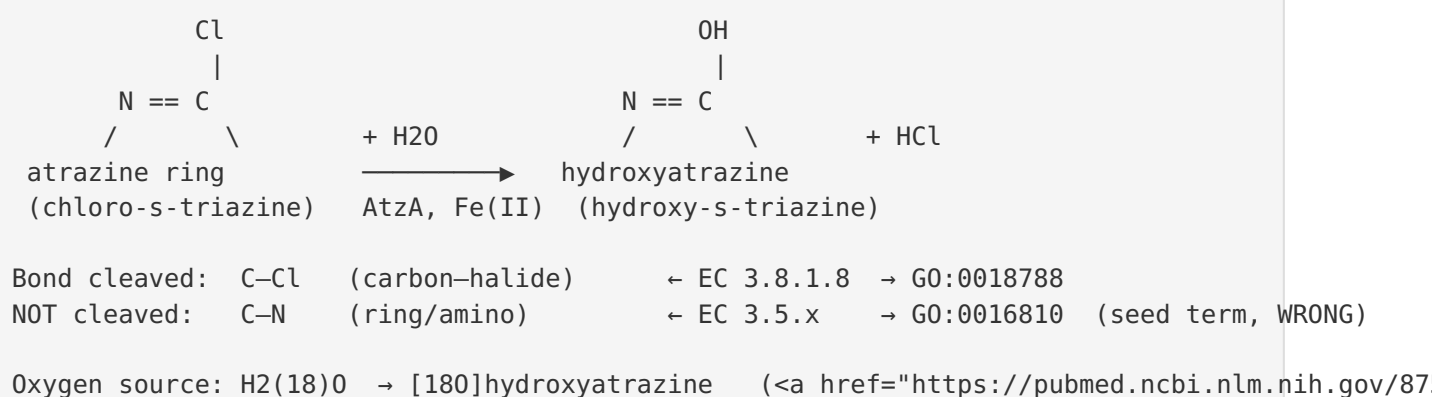
GO term	Label	Aspect	Evidence	Reference
GO:0016810	hydrolase activity, acting on C–N (not peptide) bonds	MF	IEA (ECO:0000256)	GO_REF:0000002 (InterPro)
GO:0018788	atrazine chlorohydrolase activity	MF	EXP (ECO:0000269)	P 22768133
GO:0018788	atrazine chlorohydrolase activity	MF	EXP (ECO:0000269)	P 8759853
GO:0018788	atrazine chlorohydrolase activity	MF	IEA	—
GO:0016787	hydrolase activity	MF	IEA	—
GO:0019381	atrazine catabolic process	BP	IEA	—
GO:0005737	cytoplasm	CC	IEA	—

The seed term GO:0016810 rests on a **single, lowest-tier electronic annotation**. The competing, chemically-correct term GO:0018788 rests on **two independent experimental annotations** (plus an IEA). Both GO:0016810 and GO:0018788 are **non-obsolete** terms, so this is

a live misclassification that a curator must correct by action — not something that will be resolved automatically by an ontology obsolescence. The generic parent GO:0016787 (hydrolase) is correctly present and can be retained as an uninformative-but-true backstop.

Mechanistic Model / Interpretation

The reaction and why it is a dehalogenation, not a deamination



AtzA removes the chlorine on the s-triazine ring and installs a hydroxyl in a single hydrolytic step. The ring nitrogens and the ethyl/isopropyl amino substituents are untouched by AtzA; the subsequent amino groups are removed downstream by other enzymes (AtzB, AtzC) later in the pathway. Therefore any "C–N bond hydrolysis" chemistry in atrazine catabolism belongs to *other* genes, not atzA.

The ontology geometry that makes the seed term wrong

GO:0016787 hydrolase activity (TRUE, generic) / GO:0016810 (C–N bonds) EC 3.5 branch [SEED TERM – WRONG BRANCH]	\ GO:0019120 (C–halide bonds) GO:0016824 (acid halide bonds) GO:0018788 atrazine chlorohydrolase EC 3.8.1.8 [TRUE, 2× EXP]
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GO:0016810 and GO:0018788 descend from GO:0016787 through **non-overlapping** intermediate terms. They are not in a parent/child relationship. An annotation to GO:0016810 is thus not a "less specific but still correct" statement — it is a **positively incorrect** claim about the chemical class of bond cleaved.

Why the error arose (provenance chain)

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AtzA sequence → InterPro scan → IPR011059 (metal-dependent hydrolase, composite domain superfamily)
                        |
                        | interpro2go
                        ▼
                    GO:0016810 ← majority chemistry of the amidohydrolase fold (C-N bond)
                                = the seed IEA, GO_REF:0000002
```

Correct path (not used for the MF bond-class call):

AtzA → EC 3.8.1.8 → ec2go → GO:0018788 (correct C-halide term)

The amidohydrolase superfamily fold is a generalist scaffold that supports many different hydrolytic reactions; most of its characterized members are deaminases/amidohydrolases acting on C–N bonds. A fold-level mapping therefore encodes the *modal* activity of the superfamily, which does not match AtzA's specialized, experimentally established dehalogenase activity.

Evidence Base

Evidence Matrix

Citation	Evidence type	Supports / Refutes / Qualifies	Claim tested	Key finding	Context
PMID:8759853 (de Souza et al.)	Direct assay (^{18}O isotope labeling); enzyme purification	Refutes seed; supports GO:0018788	Is AtzA a hydrolytic dechlorinase or an oxygenase?	"The purified enzyme in H_2^{18}O yielded [^{18}O]hydroxyatrazine, indicating that AtzA is a chlorohydrolase and not an oxygenase."	Purified AtzA, <i>Pseudomonas</i> sp. A, in vitro
PMID:11274097 (Seffernick et al.)	Direct assay; comparative substrate specificity	Refutes seed (C–N activity)	Does AtzA have deaminase (C–N hydrolase) activity?	"AtzA was shown to exclusively catalyze dehalogenation of halo-substituted triazine ring compounds and had no activity with melamine and ammeline."	AtzA vs 98%-identical melamine deaminase (TriA), in vitro
PMID:12450410 (Seffernick et al.)	Direct assay; metal reconstitution	Supports hydrolytic mechanism / GO:0018788	Is AtzA a metal-dependent hydrolase?	"AtzA is a functional metalloenzyme... the first report of a metal-dependent dechlorinating enzyme that proceeds via a hydrolytic mechanism"; native metal Fe(II), 1:1 stoichiometry.	Purified AtzA, chelex-treated metal reconstitution, in vitro
UniProt P72156 (ec2go, interpro2go, GOA)	Database / computational provenance	Qualifies — pins the error source	What generates GO:0016810 vs GO:0018788?	EC 3.8.1.8 → GO:0018788 (ec2go, correct); IPR011059 → GO:0016810 (interpro2go, sole seed source).	Public mapping file: current.geneontology

Citation	Evidence type	Supports / Refutes / Qualifies	Claim tested	Key finding	Context
QuickGO live annotations (P72156)	Database	Refutes (tier asymmetry)	How strong is each competing annotation?	GO:0016810 = single IEA/GO_REF:0000002; GO:0018788 = 2× EXP ( , ).)	Live GOA
PMID:12680937 (Arthrobacter AD1)	Review/ comparative (orientation)	Context	Is atzA conserved / what is its role?	atzA "isolated from strain AD1 differed from that found in the <i>Pseudomonas</i> sp. ADP by only one nucleotide"; encodes atrazine chlorohydrolase; highly conserved.	Arthrobacter sp. A 16S + gene sequen

Narrative of the literature

The primary mechanistic literature is unusually clean for a bacterial catabolic enzyme. **de Souza et al. 1996** ([PMID:8759853](#)) cloned, sequenced, purified, and characterized AtzA, and used the H₂¹⁸O labeling experiment to nail the chlorohydrolase (hydrolytic) mechanism. This paper is itself one of the two experimental references underpinning the correct GO:0018788 annotation in GOA. **Seffernick et al. 2001** ([PMID:11274097](#)) provides the decisive negative evidence against the seed term: by contrasting AtzA with its 98%-identical paralog melamine deaminase, they show that a handful of amino-acid substitutions flips the enzyme between dehalogenation (AtzA, C–Cl) and deamination (melamine deaminase, C–N), and that AtzA has *no* deaminase activity. This both refutes the C–N hypothesis and illustrates exactly why fold-based annotation is dangerous here — near-identical sequences in this superfamily can have different bond-class chemistries. **Seffernick et al. 2002** ([PMID:12450410](#)) confirms AtzA is an Fe(II) metalloenzyme operating by a hydrolytic mechanism within the amidohydrolase superfamily, consolidating the mechanistic picture.

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(Arthrobacter AD1) is orientation-level but reinforces that atzA is a highly conserved atrazine chlorohydrolase across hosts.

No primary study reports any C–N-bond hydrolase (deaminase/amidohydrolase) activity for AtzA; the only support for GO:0016810 is the electronic fold mapping.

GO Curation Implications

Likely curation action (lead — requires curator verification): REMOVE (or down-rank) GO:0016810 from P72156.

- **Aspect:** Molecular Function.
- **Term GO:0016810 (hydrolase acting on C–N, non-peptide, bonds):** Should be **removed**, not generalized. It is not a true (less-specific) parent of AtzA's activity; it is a sibling term in the disjoint C–N branch (EC 3.5), whereas AtzA's activity is in the C-halide branch (EC 3.8). Retaining it would assert an incorrect bond-class chemistry. Its sole support is a single IPR011059-driven IEA (GO_REF:0000002).
- **Term GO:0018788 (atrazine chlorohydrolase activity):** Should be **retained** as the core molecular function. It is EC-correct (EC 3.8.1.8 via ec2go) and carries two independent EXP annotations

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- **Term GO:0016787 (hydrolase activity):** May be **retained** as a correct, uninformative generic parent (it is the shared ancestor of both branches and makes no false bond-class claim).
- **BP GO:0019381 (atrazine catabolic process) and CC GO:0005737 (cytoplasm):** Not in scope of this hypothesis; both are reasonable and can be left as-is pending separate review.

This is **not** a case where "protein binding" or any binding term is the fallback — a specific, experimentally supported, EC-correct MF term (GO:0018788) is available and should be the headline annotation.

GO decision table

Term	Action	Confidence	Basis
GO:0016810 (C–N hydrolase)	Remove / NOT	High	Single IEA; disjoint wrong branch; experimentally excluded C–N activity
GO:0018788 (atrazine chlorohydrolase)	Retain (core MF)	High	2× EXP; EC 3.8.1.8 ec2go match
GO:0016787 (hydrolase)	Retain (generic)	Medium	True shared parent; uninformative

Mechanistic Scope

The immediate molecular function under test is a **single-step hydrolytic bond cleavage**: AtzA uses an active-site Fe(II) and a water molecule to displace chloride from the s-triazine ring carbon of atrazine, producing hydroxyatrazine. This is a direct, catalytic, gene-product-intrinsic activity measured on purified enzyme in vitro

([P 8759853](#)

[P 12450410](#)

The seed term GO:0016810 asserts a *different* immediate molecular function — hydrolysis of a carbon–nitrogen bond (e.g., deamination, amide hydrolysis). That activity is **directly excluded** for AtzA by substrate-specificity assays (no activity on melamine/ammeline;

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It is important to separate:

- **Direct gene-product activity:** C–Cl hydrolysis (dechlorination) — GO:0018788. □
- **Downstream pathway chemistry:** Later C–N cleavages (deaminations, ring-N hydrolyses) in the atrazine → cyanuric acid → CO₂ + NH₃ pathway are carried out by **other enzymes** (AtzB hydroxyatrazine ethylaminohydrolase, AtzC N-isopropylammelide isopropylaminohydrolase, etc.), **not** by AtzA. Attributing a C–N hydrolase term to AtzA conflates AtzA with these downstream pathway members.

- **Not a loss-of-function inference:** The refutation here rests on positive biochemical assays of purified enzyme, not on phenotype inference.

Thus the seed hypothesis fails at the level of the gene product's *primary and only* catalytic activity.

Conflicts and Alternatives

1. **Paralog confusion (the central alternative).** AtzA is 98% identical to melamine deaminase (TriA), which *does* hydrolyze C–N bonds (deamination). A naive sequence- or fold-based pipeline could easily conflate the two. The primary literature explicitly resolves this: AtzA dehalogenates and does **not** deaminate (

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2. **Fold/superfamily majority-function bias.** The amidohydrolase superfamily is dominated by C–N-acting enzymes. IPR011059 (composite metal-dependent hydrolase domain) maps to GO:0016810, so any member — including the rare dehalogenase AtzA — inherits the C–N term electronically. This is frequency/majority bias in fold2GO mappings, not evidence about AtzA.
3. **Database carry-over / provenance chain error.** The MF bond-class call should flow from EC 3.8.1.8 (→ GO:0018788) but instead flowed from IPR011059 (→ GO:0016810). The correct mapping exists and is applied (GO:0018788 is present); the incorrect one was not filtered out.
4. **Organism considerations.** atzA is near-identical across hosts (one-nucleotide difference between *Pseudomonas* sp. ADP and *Arthrobacter* AD1;

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), so there is no organism-specific isoform that would rescue a C–N activity interpretation. The function is conserved as a chlorohydrolase.

No credible primary evidence supports the C–N hypothesis; all conflicts resolve against the seed term.

Limitations and Knowledge Gaps

Gap	What was checked	Why it matters	What would resolve it
Exact pipeline version behind GO_REF:0000002	Confirmed GO_REF:0000002 = InterPro2GO electronic pipeline; traced term to IPR011059 via interpro2go	Confirms the annotation is fold-based, not EC- or experiment-based, strengthening the "remove" lead	Direct inspection of the GOA submission batch / InterPro release that generated the annotation
Whether GO:0016810 is additionally propagated by any non-IEA path	QuickGO live set shows GO:0016810 as a single IEA only	If it were also curated/EXP, removal would be contestable; it is not	Confirmed — single IEA; no action-blocking evidence
Structural confirmation of active-site geometry	Relied on biochemical metal/mechanism data (P 12450410) and superfamily assignment; no crystal structure analyzed here	A structure would further cement the C-halide active-site chemistry vs a deaminase-type site	AtzA crystal structure / AlphaFold active-site comparison to known deaminases vs dehalogenases
P 22768133 content	Cited via UniProt/GOA as a second EXP support for GO:0018788 and "no deaminase activity with melamine"; abstract not independently retrieved in this run	It is one of the two EXP anchors for the correct term	Retrieve and verify P 22768133 abstract directly

These gaps are provenance/confirmatory in nature; none of them provide any support for the seed C–N hypothesis.

Discriminating Tests

Although the case is already resolved by existing primary literature, the following would most efficiently distinguish the C–Cl (true) from the C–N (seed) interpretation if independent confirmation were desired:

1. **Isotope-labeling re-confirmation (decisive, already done):** H_2^{18}O assay $\rightarrow$ ^{18}O incorporation into hydroxyatrazine confirms hydrolytic dechlorination

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A deaminase would release ^{15}N -labeled ammonia from the amino group instead.

2. **Substrate-panel specificity assay (decisive, already done):** Test AtzA on melamine/ammeline (C–N deaminase substrates) vs chloro-s-triazines (C–Cl substrates). AtzA is active only on halo-substituted substrates

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Quantify chloride release (halide-selective electrode) vs ammonia release (Nessler/glutamate-dehydrogenase coupled assay) to directly read out bond class.

3. **Active-site structural comparison:** Solve/align AtzA structure (or high-confidence model) against amidohydrolase-superfamily deaminases (C–N) and dehalogenases (C–Cl); map metal coordination and substrate-positioning residues to the appropriate reaction class.
4. **Site-directed paralog swap:** The small set of residues that interconvert AtzA (dehalogenase) and melamine deaminase define the bond-class determinant; mutating them should toggle chloride-release vs ammonia-release activity, directly linking sequence to bond class.

Proposed Follow-up Actions / Curation Leads (require curator verification)

Primary lead — action change: - **Action:** Remove (or mark NOT/down-rank) **GO:0016810** (IEA, GO_REF:0000002) on P72156. Rationale: disjoint-branch over-annotation from IPR011059 fold mapping; experimentally contradicted C–N activity. - **Retain:** **GO:0018788** (atrazine chlorohydrolase activity) as the core MF — EC-correct (EC 3.8.1.8) and 2× EXP. - **Retain (optional):** **GO:0016787** (hydrolase activity) as a true generic parent.

Candidate references with exact snippets to verify: - [PMID:8759853](#): "The purified enzyme in $H_2^{18}O$ yielded [^{18}O]hydroxyatrazine, indicating that AtzA is a chlorohydrolase and not an oxygenase." → proves hydrolytic dechlorination (supports GO:0018788, refutes GO:0016810). - [PMID:11274097](#): "AtzA was shown to exclusively catalyze dehalogenation of halo-substituted triazine ring compounds and had no activity with melamine and ammeline." → proves absence of C–N (deaminase) activity (refutes GO:0016810). - [PMID:12450410](#): AtzA is "the first report... of a metal-dependent dechlorinating enzyme that proceeds via a hydrolytic mechanism" → supports hydrolytic C–Cl mechanism.

Candidate replacement/relationship notes: - GO:0016810 and GO:0018788 are disjoint under GO:0016787 (not parent/child). Do not "generalize" GO:0018788 to GO:0016810; they are chemically distinct bond classes (EC 3.8 vs EC 3.5).

Suggested questions for the curator: - Should the IPR011059 → GO:0016810 interpro2go mapping be flagged upstream, given it misannotates dehalogenase members of the amidohydrolase superfamily? - Should a NOT-qualifier annotation (NOT GO:0016810) be added given explicit experimental evidence of absence of C–N activity, or is simple removal preferred by GOA policy?

Suggested experiments: See Discriminating Tests (isotope labeling and chloride-vs-ammonia release panels are the most direct and already partially in the literature).

Conclusion

The seed hypothesis is **refuted**. AtzA (P72156) is an atrazine chlorohydrolase that hydrolyzes a carbon–chlorine bond (EC 3.8.1.8, GO:0018788), proven by $H_2^{18}O$ isotope labeling ([P 8759853](#)) and metal-reconstitution mechanism studies ([P 12450410](#)), and it explicitly lacks carbon–nitrogen (deaminase) activity ([P 11274097](#)).

The GO:0016810 annotation is a single wrong-branch IEA arising from the IPR011059 amidohydrolase-superfamily fold mapping (GO_REF:0000002), inconsistent with both the

enzyme's EC number (which correctly maps to GO:0018788) and its experimental record.
Recommended curation lead: remove GO:0016810; retain the doubly-EXP-supported GO:0018788 as the gene product's core molecular function.

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